

LAHTI LAUNE, Finland

WMO: 029650

Lat: 60.9622N

Lon: 25.6308E

Elev: 258

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.2	-5.9	-17.0	2.2	-12.0	-10.2	3.2	-5.8	11.6	30.8	10.0	30.0	1.0	10	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	81.5	65.7	78.1	63.9	74.9	62.5	68.4	77.0	66.3	74.6	64.3	72.1	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	94.3	72.5	63.1	87.3	69.9	61.1	81.3	67.8	32.8	77.1	31.1	74.6	29.5	72.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	9.0	7.7	-17.1	85.0	8.0	3.7	-22.9	87.7	-27.6	89.9	-32.1	92.0	-37.9	94.7		
(5)			-17.5	70.4	7.7	2.6	-23.1	72.3	-27.6	73.8	-32.0	75.2	-37.6	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	20.4	20.1	27.2	38.8	49.4	57.2	62.8	59.7	50.5	40.1	32.2	25.5		
	DBStd	16.91	11.94	12.16	8.48	6.56	7.29	5.99	5.45	5.32	5.93	7.30	8.17	10.89		
	HDD50	4586	918	837	707	341	101	8	0	3	66	312	535	758		
	HDD65	9030	1383	1257	1172	786	486	243	109	179	436	771	985	1223		
	CDD50	1097	0	0	0	5	81	225	397	302	80	7	0	0		
	CDD65	66	0	0	0	0	1	10	41	14	0	0	0	0		
	CDH74	878	0	0	0	0	50	166	482	180	0	0	0	0		
	CDH80	156	0	0	0	0	5	21	106	24	0	0	0	0		
(14) Wind	WSAvg	3.4	3.5	3.4	3.7	3.6	3.7	3.4	3.0	2.8	3.0	3.3	3.6	3.8		
(15) Precipitation	PrecAvg	23.9	1.8	1.5	1.2	1.2	1.7	2.4	2.8	2.8	2.0	2.3	2.3	1.9		
	PrecMax	31.8	3.4	3.2	2.0	2.3	3.7	4.0	7.0	5.6	5.2	5.9	5.3	4.0		
	PrecMin	18.5	0.3	0.1	0.2	0.2	0.4	0.3	0.6	0.5	0.6	0.5	0.2	0.4		
	PrecStd	3.2	0.7	0.8	0.6	0.6	0.8	1.0	1.5	1.2	1.1	1.2	1.1	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	41.3	51.4	68.0	79.5	82.9	87.0	83.1	71.6	58.9	49.5	46.0		
		MCWB	38.5	37.2	42.3	53.2	60.5	65.8	68.0	66.6	61.6	54.0	47.8	44.6		
	2%	DB	37.4	37.6	45.6	61.5	73.8	78.7	82.9	78.5	67.4	55.1	47.4	42.6		
		MCWB	36.5	36.0	38.6	48.6	58.1	62.7	67.3	64.5	58.9	52.2	46.1	41.4		
	5%	DB	35.7	35.7	41.2	56.9	69.4	75.0	79.5	75.3	64.1	53.1	45.1	40.2		
		MCWB	35.1	34.7	36.0	45.6	55.5	61.2	65.7	63.5	57.3	51.0	44.1	39.2		
	10%	DB	34.0	34.2	38.0	52.1	64.9	71.0	75.8	71.9	61.2	51.1	42.9	37.8		
		MCWB	33.3	33.4	34.6	43.4	52.8	59.1	63.9	62.1	56.3	49.5	42.0	37.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	38.1	43.5	54.2	64.4	68.8	71.7	70.0	62.3	55.3	48.5	44.5		
		MCDB	39.0	39.6	50.9	63.4	75.0	78.4	81.7	77.2	67.2	57.3	49.3	45.4		
	2%	WB	36.6	36.2	39.4	50.4	60.2	65.3	69.4	66.7	60.6	53.2	46.3	41.6		
		MCDB	37.1	37.3	44.0	59.6	69.0	74.5	79.2	74.7	65.2	54.5	47.2	42.5		
	5%	WB	35.1	34.9	37.1	47.0	57.1	62.9	67.6	64.9	58.8	51.4	44.2	39.4		
		MCDB	35.6	35.7	40.0	55.3	67.2	71.3	75.9	72.4	62.6	52.8	45.0	40.1		
	10%	WB	33.4	33.4	35.2	44.2	54.0	60.5	65.4	63.1	56.9	49.6	42.1	37.1		
		MCDB	33.9	34.1	37.3	50.9	63.2	69.0	73.4	69.8	60.4	50.9	42.8	37.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.7	15.6	18.5	21.0	19.2	19.1	19.3	16.7	11.5	7.8	8.8		
		MCDBR	7.2	8.6	20.2	28.0	29.5	27.1	25.8	26.1	21.5	13.7	8.2	7.5		
	5% WB	MCWBR	7.2	7.9	14.2	16.2	14.8	12.8	11.7	13.6	13.9	10.5	7.8	7.3		
		MCWBR	7.2	8.4	16.2	25.0	24.8	22.7	21.9	21.8	18.5	11.8	7.9	7.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.262	0.301	0.327	0.364	0.361	0.357	0.379	0.367	0.342	0.325	0.287	0.370		
		taud	2.099	2.164	2.236	2.341	2.410	2.448	2.402	2.430	2.493	2.400	2.191	2.602		
	Ebn at Noon	Ebn at Noon	146	211	249	263	274	277	266	260	247	207	136	99		
		Edh at Noon	13	23	30	32	33	32	33	30	23	18	12	8		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	75	256	725	1195	1614	1684	1636	1262	752	323	89	42		
	RadStd	RadStd	14	56	92	125	177	130	158	140	98	55	16	6		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		+0.88	N/A	N/A	N/A	N/A	N/A	-266	-299	N/A	N/A	

Nomenclature: See separate page